THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME — SET 1

Chapter 7: **Probability** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	(d) $\frac{11}{50}$	Relative frequency of the outcome.
2	(c) $\frac{7}{25}$	Relative frequency.
3	(c) $\frac{5}{9}$	Count vowels among all the letters (repeats counted).
4	(d) $\frac{3}{8}$	Use the 8 equally likely outcomes.
5	(a) $\frac{3}{4}$	Count the balls that are not black.
6	(b) 70	$P(\text{not } E) = 1 - P(E)$.
7	(b) $\frac{1}{3}$	Count favourable outcomes out of 6.
8	(b) $0 \le P(E) \le 1$	Recall the definition.
9	(b) $\frac{7}{50}$	Relative frequency of the outcome.
10	(b) $\frac{9}{40}$	Relative frequency.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Even outcomes: $2, 4, 6$ with frequencies $22, 16, 16$ (1); $P = \frac{54}{100} = \frac{27}{50}$ (1).	1+1
12	Number of letters $= 11$ (1); $P = \frac{7}{11}$ (1).	1+1
13	Sample space has 8 outcomes (1); $P = \frac{3}{8}$ (1).	1+1
14	Even outcomes: $2, 4, 6$ with frequencies $12, 15, 16$ (1); $P = \frac{43}{100} = \frac{43}{100}$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Total $= 12$ (1); $P(\text{red}) = \frac{1}{3}$ (1); $P(\text{not red}) = \frac{2}{3}$; $P(\text{black or white}) = \frac{2}{3}$ (1).	1+1+1
16	Walk $= 200 - 80 - 50 = 70$ (1); $P(\text{walk}) = \frac{7}{20}$ (1); $P(\text{not bus}) = 1 - \frac{2}{5} = \frac{3}{5}$ (1).	1+1+1
17	$P(2 \text{ heads}) = \frac{24}{100} = \frac{6}{25}$ (1); $P(1 \text{ head}) = \frac{52}{100} = \frac{13}{25}$ (1); $P(0 \text{ heads}) = \frac{24}{100} = \frac{6}{25}$ (1).	1+1+1
	OR: Total $= 15$ (1); $P(\text{red}) = \frac{2}{5}$ (1); $P(\text{not red}) = \frac{3}{5}$; $P(\text{black or white}) = \frac{3}{5}$ (1).	1+1+1
18	Walk $= 250 - 100 - 75 = 75$ (1); $P(\text{walk}) = \frac{3}{10}$ (1); $P(\text{not bus}) = 1 - \frac{2}{5} = \frac{3}{5}$ (1).	1+1+1
	OR: $P(2 \text{ heads}) = \frac{45}{200} = \frac{9}{40}$ (1); $P(1 \text{ head}) = \frac{105}{200} = \frac{21}{40}$ (1); $P(0 \text{ heads}) = \frac{50}{200} = \frac{1}{4}$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Total trials $= 100$ (1); Prime: $2, 3, 5$: $\frac{53}{100} = \frac{53}{100}$ (1); Greater than 4: $\frac{30}{100} = \frac{3}{10}$ (1); Less than 3: $\frac{37}{100} = \frac{37}{100}$ (1); All six frequencies add to 100, so relative frequencies add to 1 (1).	1+1+1+1+1
	OR: Number of letters $= 11$ (1); $P(\text{vowel}) = \frac{4}{11}$ (1); $P(S) = \frac{4}{11}$ (1); $P(A \text{ to } M) = \frac{5}{11}$ (1); Repeated letters are counted each time (1).	1+1+1+1+1
20	Total trials $= 100$ (1); Prime: $2, 3, 5$: $\frac{51}{100} = \frac{51}{100}$ (1); Greater than 4: $\frac{35}{100} = \frac{7}{20}$ (1); Less than 3: $\frac{30}{100} = \frac{3}{10}$ (1); All six frequencies add to 100, so relative frequencies add to 1 (1).	1+1+1+1+1
	OR: Number of letters $= 11$ (1); $P(\text{vowel}) = \frac{4}{11}$ (1); $P(M) = \frac{2}{11}$ (1); $P(A \text{ to } M) = \frac{8}{11}$ (1); Repeated letters are counted each time (1).	1+1+1+1+1

Mistakes to avoid:

- Giving a probability greater than 1 or a negative value, or leaving it as a count instead of a fraction.
- Using the number of favourable outcomes divided by the number of *unfavourable* outcomes instead of the *total* outcomes.
- Listing only HT for 'one head' when two coins are tossed; HT and TH are different outcomes.
- Treating 'at least one' as 'exactly one'; use the complement $1 - P(\text{none})$.
- Assuming that outcomes are always equally likely, e.g. 'the sum of two dice is 7' has probability $\frac{6}{36}$, not $\frac{1}{11}$.
- Confusing experimental and theoretical probability: experimental values come from trials and may differ from $\frac{1}{2}$ for a coin.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 7: **Probability** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** $\frac{1}{3}$	Count vowels among all the letters (repeats counted).
2	**(a)** $\frac{1}{8}$	Use the 8 equally likely outcomes.
3	**(b)** $\frac{13}{15}$	Count the balls that are not black.
4	**(d)** 60	$P(\text{not } E) = 1 - P(E)$.
5	**(a)** $\frac{1}{3}$	Count favourable outcomes out of 6.
6	**(a)** equally likely	Recall the definition.
7	**(b)** $\frac{3}{25}$	Relative frequency of the outcome.
8	**(b)** $\frac{3}{10}$	Relative frequency.
9	**(c)** $\frac{4}{11}$	Count vowels among all the letters (repeats counted).
10	**(b)** $\frac{1}{2}$	Use the 8 equally likely outcomes.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Number of letters $= 9$ (1); $P = \frac{1}{3}$ (1).	1+1
12	Sample space has 8 outcomes (1); $P = \frac{1}{2}$ (1).	1+1
13	Even outcomes: $2, 4, 6$ with frequencies $17, 20, 15$ (1); $P = \frac{52}{100} = \frac{13}{25}$ (1).	1+1
14	Number of letters $= 11$ (1); $P = \frac{7}{11}$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Total $= 15$ (1); $P(\text{red}) = \frac{1}{3}$ (1); $P(\text{not red}) = \frac{2}{3}$; $P(\text{black or white}) = \frac{2}{3}$ (1).	1+1+1
16	Walk $= 120 - 45 - 30 = 45$ (1); $P(\text{walk}) = \frac{3}{8}$ (1); $P(\text{not bus}) = 1 - \frac{3}{8} = \frac{5}{8}$ (1).	1+1+1
17	$P(2\text{ heads}) = \frac{18}{80} = \frac{9}{40}$ (1); $P(1\text{ head}) = \frac{42}{80} = \frac{21}{40}$ (1); $P(0\text{ heads}) = \frac{20}{80} = \frac{1}{4}$ (1).	1+1+1
	OR: Total $= 20$ (1); $P(\text{red}) = \frac{7}{20}$ (1); $P(\text{not red}) = \frac{13}{20}$; $P(\text{black or white}) = \frac{13}{20}$ (1).	1+1+1
18	Walk $= 300 - 120 - 90 = 90$ (1); $P(\text{walk}) = \frac{3}{10}$ (1); $P(\text{not bus}) = 1 - \frac{2}{5} = \frac{3}{5}$ (1).	1+1+1
	OR: $P(2\text{ heads}) = \frac{38}{160} = \frac{19}{80}$ (1); $P(1\text{ head}) = \frac{78}{160} = \frac{39}{80}$ (1); $P(0\text{ heads}) = \frac{44}{160} = \frac{11}{40}$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Total trials $= 100$ (1); Prime: $2, 3, 5$: $\frac{44}{100} = \frac{11}{25}$ (1); Greater than 4: $\frac{28}{100} = \frac{7}{25}$ (1); Less than 3: $\frac{38}{100} = \frac{19}{50}$ (1); All six frequencies add to 100, so relative frequencies add to 1 (1).	1+1+1+1+1
	OR: Number of letters $= 11$ (1); $P(\text{vowel}) = \frac{6}{11}$ (1); $P(N) = \frac{2}{11}$ (1); $P(A \text{ to } M) = \frac{6}{11}$ (1); Repeated letters are counted each time (1).	1+1+1+1+1
20	Total trials $= 120$ (1); Prime: $2, 3, 5$: $\frac{58}{120} = \frac{29}{60}$ (1); Greater than 4: $\frac{40}{120} = \frac{1}{3}$ (1); Less than 3: $\frac{40}{120} = \frac{1}{3}$ (1); All six frequencies add to 120, so relative frequencies add to 1 (1).	1+1+1+1+1
	OR: Number of letters $= 11$ (1); $P(\text{vowel}) = \frac{4}{11}$ (1); $P(B) = \frac{2}{11}$ (1); $P(A \text{ to } M) = \frac{6}{11}$ (1); Repeated letters are counted each time (1).	1+1+1+1+1

Mistakes to avoid:

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- Listing only HT for 'one head' when two coins are tossed; HT and TH are different outcomes.
- Treating 'at least one' as 'exactly one'; use the complement $1 - P(\text{none})$.
- Assuming that outcomes are always equally likely, e.g. 'the sum of two dice is 7' has probability $\frac{6}{36}$, not $\frac{1}{11}$.
- Confusing experimental and theoretical probability: experimental values come from trials and may differ from $\frac{1}{2}$ for a coin.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 7: **Probability** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** $\frac{2}{3}$	Count the balls that are not black.
2	**(b)** 40	$P(\text{not } E) = 1 - P(E)$.
3	**(c)** $\frac{2}{3}$	Count favourable outcomes out of 6.
4	**(a)** sample space	Recall the definition.
5	**(a)** $\frac{19}{100}$	Relative frequency of the outcome.
6	**(b)** $\frac{7}{20}$	Relative frequency.
7	**(a)** $\frac{3}{10}$	Count vowels among all the letters (repeats counted).
8	**(d)** $\frac{3}{8}$	Use the 8 equally likely outcomes.
9	**(c)** $\frac{11}{20}$	Count the balls that are not black.
10	**(b)** 30	$P(\text{not } E) = 1 - P(E)$.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Sample space has 8 outcomes (1); $P = \frac{1}{8}$ (1).	1+1
12	Number of letters $= 9$ (1); $P = \frac{5}{9}$ (1).	1+1
13	Number of letters $= 11$ (1); $P = \frac{5}{11}$ (1).	1+1
14	Sample space has 8 outcomes (1); $P = \frac{3}{8}$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Total $= 15$ (1); $P(\text{red}) = \frac{1}{5}$ (1); $P(\text{not red}) = \frac{4}{5}$; $P(\text{black or white}) = \frac{4}{5}$ (1).	1+1+1
16	Walk $= 180 - 60 - 40 = 80$ (1); $P(\text{walk}) = \frac{4}{9}$ (1); $P(\text{not bus}) = 1 - \frac{1}{3} = \frac{2}{3}$ (1).	1+1+1
17	$P(2\text{ heads}) = \frac{72}{300} = \frac{6}{25}$ (1); $P(1\text{ head}) = \frac{150}{300} = \frac{1}{2}$ (1); $P(0\text{ heads}) = \frac{78}{300} = \frac{13}{50}$ (1).	1+1+1
	OR: Total $= 15$ (1); $P(\text{red}) = \frac{2}{15}$ (1); $P(\text{not red}) = \frac{13}{15}$; $P(\text{black or white}) = \frac{13}{15}$ (1).	1+1+1
18	Walk $= 400 - 150 - 100 = 150$ (1); $P(\text{walk}) = \frac{3}{8}$ (1); $P(\text{not bus}) = 1 - \frac{3}{8} = \frac{5}{8}$ (1).	1+1+1
	OR: $P(2\text{ heads}) = \frac{12}{50} = \frac{6}{25}$ (1); $P(1\text{ head}) = \frac{27}{50} = \frac{27}{50}$ (1); $P(0\text{ heads}) = \frac{11}{50} = \frac{11}{50}$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Total trials $= 120$ (1); Prime: $2, 3, 5$: $\frac{53}{120} = \frac{53}{120}$ (1); Greater than 4: $\frac{39}{120} = \frac{13}{40}$ (1); Less than 3: $\frac{43}{120} = \frac{43}{120}$ (1); All six frequencies add to 120, so relative frequencies add to 1 (1).	1+1+1+1+1
	OR: Number of letters $= 10$ (1); $P(\text{vowel}) = \frac{3}{10}$ (1); $P(H) = \frac{1}{5}$ (1); $P(A \text{ to } M) = \frac{4}{5}$ (1); Repeated letters are counted each time (1).	1+1+1+1+1
20	Total trials $= 120$ (1); Prime: $2, 3, 5$: $\frac{64}{120} = \frac{8}{15}$ (1); Greater than 4: $\frac{41}{120} = \frac{41}{120}$ (1); Less than 3: $\frac{44}{120} = \frac{11}{30}$ (1); All six frequencies add to 120, so relative frequencies add to 1 (1).	1+1+1+1+1
	OR: Number of letters $= 9$ (1); $P(\text{vowel}) = \frac{1}{3}$ (1); $P(A) = \frac{1}{3}$ (1); $P(A \text{ to } M) = \frac{5}{9}$ (1); Repeated letters are counted each time (1).	1+1+1+1+1

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- Assuming that outcomes are always equally likely, e.g. 'the sum of two dice is 7' has probability $\frac{6}{36}$, not $\frac{1}{11}$.
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